MILITARY SPECIFICATION

# SAFE AND ARMING (PARTIAL LOADING) ASSEMBLY FOR MINES, M718/M741 SYSTEM

This Specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This Specification contains requirements not normally covered by the drawings, and provides Quality Assurance provisions for the assembly of a safe and arming device for mines used in the M718/M741 System ballistic tests.

## 2. APPLICABLE DOCUMENTS

2.1 *Issue of documents*. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | | |
|---|---|---|
| MIL-A-48078 | - | Ammunition, Standard Quality Assurance Provisions General Specification For. |
| MIL-D-48120 | - | Detonator, Electric: XM100; Parts for and Loading, Assembling and Packing. |
| MIL-D-63197 | - | Detonator Delay Electric: PA506; Parts For, and Loading, Assembling and Packing. |
| MIL-L-63246 | - | Lead Assemblies for S&A Mechanism for Scatterable Mines; Parts for, and Loading, Assembling and Packing. |

PSC: 1345

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

| | | |
|---|---|---|
| MIL-STD-105 | - | Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105) |
| MIL-STD-129 | - | Marking for Shipment and Storage |
| MIL-STD-453 | - | Inspection, Radiographic |

DRAWINGS (See 6.7)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

| | | |
|---|---|---|
| 9317603 | - | Safe and Arming (Partial Loading) Assembly |

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

| | | |
|---|---|---|
| Title 49 | - | Transportation, Parts 100-199 |

(The Code of Federal Regulations if available from the Superintendent of Documents, U.S. Government Printing Officer, Washington, DC 20402. Orders should specify "49 CFR 0-190 (lastest revision)").

(Copies of specifications, standards, drawings, and publications listed above, or in the item detail specification, required by suppliers in connection with specification procurement functions, should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts and assemblies. The parts and assemblies shall comply with all requirements specified on drawings listed in paragraph 2.1, and associated drawings, and with all requirements specified in applicable specifications and standards.



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3.3 Radiographic examination. The safe and arming device shall be examined for a fully non-armed condition in accordance with MIL-STD-453 and 4.5.4 and in accordance to Dwg. 9317603. Further radiographic requirements shall include clear definition of all explosive components as well as flexible cable assembly and slider lock. Verify that center PA513 (Dwg. 9278002) and main PA514 (Dwg. 9277795) explosive leads are not present.

3.4 Detonator short circuit resistance. The resistance of the short circuit across the detonators on the flexible cable shall be less than fifteen thousandth (0.015) ohms when tested in accordance with 4.5.5.2.

3.5 Explosive functioning. The detonators and transfer lead shall all function high order when the detonators are initiated in accordance with the test procedure of 4.5.8.

3.6 Data requirements. The contractor shall forward copies of the results of any of his examinations or tests to the designated address, as required by the contract (See 6.5).

3.7 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.8 Workmanship - parts and assemblies. Parts and assemblies shall be devoid of burrs, chips, dirt, grease, and other foreign matter that may impair proper functioning. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated. Exterior surface coatings shall be continuous, except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection and standard quality assurance provisions. Unless otherwise specified herein, or in the contract, the provisions of MIL-A-48078 shall apply, and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item.

a. First Article Inspection

b. Quality Conformance Inspection



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4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting Officer, for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated:

| Part Description | Drawing | Quantity |
|---|---|---|
| Slider Assembly | 9317618 | 10 |
| Safe and Arming (Partial Loading) Subassembly | 9317603 | 10 |
| Cable and Detonator Assembly | 9277868 | 10 |
| Safe and Arming (Partial Loading) Assembly | 9317603 | 35 |
| Setback Pin Assembly | 9317617 | 10 |

4.3.2 Inspection to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provision of MIL-A-48078. In addition, inspection lots of safe and arming assemblies & sub-assemblies shall contain:

a. Parts bearing the same lot interfix number from one manufacturer.

b. Explosive detonators, leads from the same lot interfix number from one manufacturer.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105.



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TABLE I - FIRST ARTICLE INSPECTION

## CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63382(AR)

| PARAGRAPH | TITLE: Safe and Arming Device (Partial Loading) for M718/M741 System | | SHEET 1 OF 2 | | DRAWING NUMBER: See Below |
|---|---|---|---|---|---|
| | | | | | NEXT HIGHER ASSEMBLY |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| | Slider Assembly (Dwg. No. 9317618) | | | | |
| | Examination for defects | 10 | | 3.2 | 4.4.2.1 |
| | Safe and Arming Partial Loading Subassembly (Dwg. No. 9317603) | | | | |
| | Examination for defects | 10 | | 3.2 | 4.4.2.2 |
| | Non-Arming Test | 10 | | 3.2 | 4.5.1 |
| | Non-Arming Test of centrifugal weights | 10 | | 3.2 | 4.5.3 |
| | Arming Test | 10 | | 3.2 | 4.5.2 |
| | Cable and Detonator Assembly (Dwg. No. 9277868) | | | | |
| | Examination for defects | 10 | | 3.2 | 4.4.2.3 |
| | Safe and Arming (Partial Loading) Assembly (Dwg. No. 9317603) | | | | |
| | Examination for defects | 10 | | 3.2 | 4.4.2.4 |
| | Tests | | | | |
| | Detonator Resistance | 25 | | 3.2 | 4.5.5.1 |
| | Detonator Short Circuit | 25 | | 3.4 | 4.5.5.2 |
| | Closure, Firing Pin Stake Push Off Test | 5 | | 3.2 | 4.5.6 |
| | Closure, Slider Stake Push Off Test | 5 | | 3.2 | 4.5.6 |
| | Radiographic Examination | 10 | | 3.3 | 4.5.4 |
| | Explosive Test (see para. 6.8) | 25 | | 3.5 | 4.5.8 |

NOTES:

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TABLE I - FIRST ARTICLE INSPECTION

## CLASSIFICATION OF DEFECTS & TESTS

| PARAGRAPH | TITLE | | SHEET 2 OF 2 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| | Safe and Arming Device (Partial Loading) for M718/M741 System | | | | See Below |
| | | | | | NEXT HIGHER ASSEMBLY |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| | Setback Pin Assembly (Dwg. No. 9317617) Examination for defects | 10 | | 3.2 | 4.4.2.5 |

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QUALITY CONFORMANCE INSPECTION

# CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63382(AR)

| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.1 | Slider Assembly | 1 OF 1 | 9317618 |
| | | | NEXT HIGHER ASSEMBLY 9317603 |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Center charge lead assembly slug missing | | 100% | 3.2 | Visual |
| 102 | Transfer charge lead assembly missing, or assembled incorrectly | | 100% | 3.2 | Visual |
| 103 | Groove pin missing, or assembled incorrectly | | 100% | 3.2 | Visual |
| 104 | Lead holder transfer charge stake improper | | 100% | 3.2 | Visual |
| 105 | Lead holder missing | | 100% | 3.2 | Visual |
| Minor | | | | | |
| 201 | Evidence of poor workmanship | | 1.0% | 3.8 | Visual |

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QUALITY CONFORMANCE INSPECTION

## CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63382(AR)

| PARAGRAPH | TITLE | | SHEET 1 OF 2 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.2 | Safe and Arming (Partial Loading) Subassembly | | | | 9317603(3) |
| | | | | | NEXT HIGHER ASSEMBLY |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | | | | | |
| 1 | Spring, centrifugal lock missing | | 100%(1) | 3.2 | Visual/Manual |
| 2 | Holder, spring, missing or loose | | 100% | 3.2 | Visual/Manual |
| 3 | Lock, centrifugal missing | | 100% | 3.2 | Visual |
| 4 | Any spring missing or improperly positioned | | 100% | 3.2 | Visual/Manual |
| 5 | Presence of "G" weight | | 100% | 3.2 | Visual/Manual |
| 6 | Presence of ball | | 100% | 3.2 | Visual/Manual |
| 7 | Presence of "G" weight lock | | 100% | 3.2 | Visual/Manual |
| 8 | Presence of set back pin assembly | | 100% | 3.2 | Visual/Manual |
| 9 | Non-arming test | | 100% | 3.2 | 4.5.1 |
| 10 | Non-arming centrifugal lock test | | 100%(4) | 3.2 | 4.5.3 |
| Major | | | | | |
| 101 | Presence of more than one spring per cavity | | 100% | 3.2 | Visual/Manual |
| 102 | Staking missing or improper | | 100% | 3.2 | Visual |
| 103 | Presence of rear locks | | 100% | 3.2 | Visual |
| 104 | Presence of rear lock springs | | 100% | 3.2 | Visual |
| 105 | Any closure missing, loose or improperly staked | | 100% | 3.2 | Visual/Manual |
| 106 | Arming test | | 100%(2)(4) | 3.2 | 4.5.2 |

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QUALITY CONFORMANCE INSPECTION

## CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63382(AR)

| PARAGRAPH | TITLE | | SHEET 2 OF 2 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.2 | Safe and Arming (Partial Loading) Subassembly | | | | 9317603(3) |
| | | | | | NEXT HIGHER ASSEMBLY |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Minor 201 | Evidence of poor workmanship<br><br>Notes:<br>(1) Manual inspection is defined as the allowable probing of the unit by means of an approved instrument which will not cause damage to the sample.<br>(2) Test is to be performed prior to inserting rear lock assembly.<br>(3) Parts list shall be in accordance with 4.5.1.<br>(4) Arming test shall always precede the non-arming tests. | | 1.00% | 3.8 | Visual |

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QUALITY CONFORMANCE INSPECTION

## CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63382(AR)

| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.3 | Cable and Detonator Assembly | 1 OF 1 | 9277868 |
| | | | NEXT HIGHER ASSEMBLY: 9317603 |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Solder improper | | 100% | 3.2 | Visual |
| 102 | Detonator placement incorrect | | 0.40% | 3.2 | Visual |
| 103 | Assembly height, maximum | | 0.40% | 3.2 | Gage |
| Minor | | | | | |
| 201 | Evidence of poor workmanship | | 1.0% | 3.8 | Visual |

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QUALITY CONFORMANCE INSPECTION.

## CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63382(AR)

| PARAGRAPH | TITLE | SHEET 1 OF 1 | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.4 | Safe and Arming Assembly (Partial Loading) Assembly | | 9317603 |
| | | | NEXT HIGHER ASSEMBLY |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| Critical | | | | | |
| 1 | Assembly in partially armed or armed state or missing components or improper explosive loading(radiographic) | | 100% | 3.3 | 4.5.4 |
| Major | | | | | |
| 101 | Grounding clip missing or positioned improperly | | 100% | 3.2 | Visual |
| 102 | Cable and detonator assembly not fully seated | | 100% | 3.2 | Visual |
| 103 | Solder improper (grounding clip tab to cable assembly) | | 100% | 3.2 | Visual/Manual |
| 104 | Presence of all closure discs | | 100% | 3.2 | Visual/Manual |
| 105 | Shield assembly missing, loose, not staked, or improperly installed | | 100% | 3.2 | Visual/Manual |
| 106 | Detonator resistance (After shield assy) | | 100% | 3.2 | 4.5.5.1 |
| 107 | Detonator not shorted (After shield assy) | | 100%(1) | 3.4 | 4.5.5.2 |
| 108 | Cable spacer missing or improperly installed | | 100%(2) | 3.2 | Visual |
| 109 | Closure firing pin stake push off test | --- | --- | 3.2 | 4.4.3.1/4.5.6 |
| 110 | Closure, slider stake push off test | --- | --- | 3.2 | 4.4.3.1/4.5.6 |
| 111 | Explosive Test (see para. 6.8) | | 0.40% | 3.5 | 4.5.8 |
| Minor | | | | | |
| 201 | Evidence of poor workmanship | | 1.0% | 3.8 | Visual |

NOTES: (1) Verify short circuit after resistance measurements have been taken and the short circuit has been replaced
(2) Inspection must be made during assembly operation.

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## CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63382(AR)

| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.5 | Setback Pin Assembly | | | | 9317617 |
| | | | | | NEXT HIGHER ASSEMBLY |
| | | | | | 9317603 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS. | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | None defined | | | | |
| Major 101 | Protrusion of stop pin | | 0.40% | 3.2 | Gage |
| Minor 201 | Evidence of poor workmanship | | 1.0% | 3.8 | Visual |

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables and below.

4.4.3.1 Stake push off test. A "Push off test" shall be performed on the front and rear housing closures after being staked to the housing, to check for integrity of staking. The test shall be performed on two samples each, prior to the start of each shift, and then performed on one sample each, after every two hours production and at the end of the shift, until three consecutive shifts have been accepted. Thereafter, the test may be performed on one sample each, prior to the start of each shift, and one sample each after four hours production and at the end of the shift. Failure of a test sample at any time will cause the test to revert to the initial test conditions until three consecutive shifts have been accepted.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is identified in the tables beginning with paragraph 4.4.2.1. The contractor shall submit inspection equipment designs for approval, in accordance with the terms of the contract. See Section 6 of MIL-A-48078, and Section 6.3 herein.

4.5 Test methods and procedures

4.5.1 Non-Arming of safe and arming subassembly. The subassembly shall be tested for the specified non-arming requirements using the following parts as a minimum for the sample.

| Part No. | Nomenclature |
|---|---|
| 9277786 | Centrifugal locks |
| 9277790 | G-Weight |
| 9277871 | G-Weight lock |
| 9278008 | Housing |
| 9317618 | Slider assembly |
| 9317617 | Set back pin assembly |
| MS19060-4810 | Ball bearing |

Also included are all associated springs, locks and closures for the above assembly.

Optional parts are:

Closure firing pin, Part No. 9317591
Rear locks, Part No. 9278057
Slider, lock, Part No. 9278060
Rear lock springs, Part No. 9278109
And all associated closures.



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Also included for the optional parts are all associated springs, locks and closures for the above assembly.

The test sample must not contain the staked disc closure, Part No. 9277812 for this test.

The slider shall be maintained in the correctly oriented position during the arming and non-arming tests. The Contractor, when necessary, may utilize an interim slider lock for test purposes.

The S&A assembly shall be tested as follows using approved spin and testing equipment.

The S&A, after secured nesting in the spin fixture, shall be spun up to the arming speed specified on the applicable drawing.

To verify that the set back pin is in the non-armed position, the contractor shall attempt to arm the "G" weight (Dwg 9277790), by applying the minimum arming force as stated on the applicable drawing. The S&A sub-assembly shall not arm. The force on the "G" weight shall then be released.

The set back pin assembly (Dwg 9317617) shall be moved to the armed position and held in place. Arming force shall be as specified in drawing 9317603.

The S&A, shall be spun to the non-arming speed as specified on the applicable drawing. The set back pin shall remain in the armed position. The contractor shall attempt to arm the "G" weight (Dwg 9277790), in accordance with the minimum all-arm requirement as stated on the applicable drawing.

While complying with the above, neither the G-weight, Part No. 9277790, the ball bearing, Part No. MS-19060-4810 or the slider assembly, Part No. 9317618 shall move out of the safe state, i.e., into the armed position or in a position resting on the centrifugal locks. Duration of the time of spin for the test equipment shall also be approved by the technical agency of 6.5.

After or during this spin test, the sub-assembly shall be examined for evidence of arming either in or out of the test nest.

Any S&A failing to comply with any or all of the non-arming requirements shall be classed defective and removed from the lot. (Non-destructive test).



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4.5.2 Arming of safe and arming sub-assembly. The subassembly, consisting of the parts specified in 4.5.1 shall be spun up to the arming speed specified in Dwg. 9317603 using approved test equipment. When at the required arming speed and the set back pin assembly, Part No. 9317617, in the armed position, the G-weight part No. 9277790, shall be moved into the armed position and the slider assembly, Part No. 9317618, shall move into the full arm position. Failure of the sample to comply with all of the specified requirements shall cause the sample to be classed defective and removed from the lot.

NOTE: Arming forces on the G-Weight, Dwg. No. 9277790, and Setback Pin Assembly, Dwg. No. 9317617, need not be rigoriously controlled or measured during spin if it can be shown that the static force requirements are met.

The sample shall be disarmed at the end of the test by entering into further tests (See 4.5.3) or by other means as approved by the technical agency listed in 6.5. (Non-destructive test)

4.5.3 Non-arming of centrifugal locks (safe and arming sub-assembly). An armed test sample, using approved test equipment shall be spun at the non-arming speed specified in Dwg. 9317603. While at this speed, the slider assembly, part No. 9317618, shall be moved into the non-armed position and the instrument performing this operation shall be withdrawn or removed. The slider assembly, without any physical restraint upon it shall not move into the full armed position but must remain in a non-armed position. This shall be indicated by the slider assembly coming to rest on top of the centrifugal locks Part No. 9277786.

The S&A shall be in the full unarmed (safe) position after all tests have been completed. With the exception of the set back pin non-arming test specified in paragraph 4.5.1, (which may be performed during the arming sequence for convenience) all non-arming tests shall be performed after the arming tests. The order of the tests within the above arming and non-arming paragraphs is optional. Any S&A failing to comply with any or all of the requirements shall be classed defective and removed from the lot. (Non-destructive tests).

4.5.4 Radiographic examination. - The safe and arming assemblies radiographic plates shall be of a quality such that the Technical Agency of 6.5 or their appointed representatives shall be able to view all of the requirements to the Agency's satisfaction. The radiographic plates shall clearly define all explosive components; positively show that the assembly is in the full non-armed condition; the flexible cable shall not be pinched, twisted or damaged; and all components shall be in the non-armed position. Any assembly failing to comply with all of the specified requirements shall be classed defective and removed from the lot. (Non-destructive test).



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4.5.5 Resistance measurements

4.5.5.1 Detonator continuity resistance. The resistance of the detonators as installed on the flexible cable and safe and arming assembly shall be measured for the value specified in (Dwg. 9317603). An approved millohmmeter whose excitation current does not exceed the no fire condition of the detonators shall be used. The resistance measurements shall be obtained across the eyelets on the flexible cable (Dwg. 9277774). Any assembly exhibiting resistance not conforming to the specified requirement shall be classed defective. The exposed conductor in the flexible cable (Dwg 9277774) must be cut to perform this test and soldered to reconnect the above conductor after completion of test. (Non-destructive test).

Any assembly failing to comply with the specified resistance requirements shall be classed defective. (Non-destructive test.)

4.5.5.2 Detonator short circuit resistance. The soldered flexible cable assembly shall be measured for the specified resistance using an approved milliohmmeter whose excitation current does not exceed the no fire condition of detonators across the eyelets. This test must be performed after the device passes the conditions specified in 4.5.5.1.

Any assembly failing to comply with the specified resistance requirements shall be classed defective. (Non-destructive test.)

4.5.6 Stake push off test. The test shall be performed using approved test equipment on samples of housings and closures staked in accordance with the specified requirements and with no components within the housing. If the sample fails to comply with all of the specified requirements it shall be classed defective and all production items preceding the last successful push test shall be rejected. (Destructive test).

4.5.7 Check test for deterioration of detonators and leads. (See MIL-L-63246, MIL-D-63197 and MIL-D-48120). If the total time between original acceptance of any detonator or lead lot and the assembly of the explosive device lot into the safe and arming device exceeds two years, or if the detonators or leads have been subjected to adverse conditions, however brief, at any time since previous tests, the detonator or lead lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in MIL-L-63246 and/or MIL-D-63197, and/or MIL-D-48120. Immediately before the detonator or lead lot is assembled, the Government Inspector at the facility assembling the detonators or leads into the safe and arming device shall select the sample for deterioration tests and the contractor shall verify that the devices comply with the requirements of the above specifications.



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4.5.8 Explosive train test. The armed safe and arming device shall be tested in an explosion proof test chamber. (To accomplish the test below the soldered short on the flexible cable must be removed.) An eighty three (83) microfarad plus or minus five percent (± 5%) capacitor charged to five plus or minus one tenth (5.0 ± 0.1) volts direct current shall be applied to the cable inputs which will cause the assembly to explosively function. A visual examination shall be made of the assembly to ascertain if all components have functioned. Any sample failing to comply with all of the specified test requirements shall be classed defective. (Destructive test)

NOTE 1: The explosive train performance test may be performed on assemblies consisting of all explosive elements, housings, slider assemblies, slider locks and staked shield assembly assembled in the unarmed condition, and then armed remotely. The contractor shall describe the proposed method of remote arming (see para. 4.4.4). All explosive elements and metal parts shall be a representative sample drawn from each lot of parts that will constitute one lot of safe and arming assemblies (i.e.; complying with the lotting provisions of 4.4.1).

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level - Commerical. The safe and arming device shall be preserved and packaged utilizing industrial procedures and in accordance with Code of Federal Regulations Title 49 100-199. These procedures shall be submitted for examination in accordance with Para. 6.5.

5.2 Packing

5.2.1 Level - Commerical. The safe and arming device preserved and packaged as specified in 5.1.1 shall be packed for shipment in accordance with acceptable industrial procedures such as to prevent deterioration or damage during handling and shipment to the first destination. Industrial packing procedures shall be in accordance with Code of Federal Regulations Title 49 100-199

5.3 Marking. Marking shall be in accordance with MIL-STD-129 and Code of Federal Regulations Title 49 100-199.

5.4 Shipping. When shipments from more than one lot are shipped, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.



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6. NOTES

6.1 Intended use. The assembly covered by this specification is intended for use in the Projectile, 155mm, AT, M718/M741 as a special ballistic test item.

6.2 Ordering data. See MIL-STD-48078.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, ARRADCOM, Attn: DRDAR-QAA-I, Dover, New Jersey 07801.

6.4 Cost of check test. The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test for deterioration of the detonators and/or leads. The tests shall be conducted at Government expense without cost to the contractor who loaded the detonators and/or leads or to the contractor assembling them into the Safe and arming device. This shall not constitute a basis for rejection against either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., while the lot of detonators and/or leads was under the jurisdiction of either contractor.

6.5 Submission of results of contractor conducted examinations and tests. Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commander, ARRADCOM, Attn: DRDAR-QAA-M.

6.6 Distribution of ammunition data cards. Distribution of data cards shall include the following: Commander, ARRADCOM, Attn: DRDAR-QAA-M, Dover, New Jersey 07801.

6.7 Drawings. Drawings listed in Section 2 of this Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities are now under cognizance of ARRADCOM.

6.8 Explosive test. The contractor may request a waiver of the explosive test provided all of the following criteria are met:

(a) The contractor is currently producing Safe and Arming Devices for Mines, M718/M741 Systems, P/N 9278015.



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(b) The metal parts, detonators, cable and detonator assemblies and transfer lead are from the same lots of components used to produce a lot of P/N 9278015 S&A's.

(c) The above lot of 9278015 S&A's successfully passes the explosive train test specified in MIL-S-48747A.

Custodian:
Army-AR

Preparing Activity:
Army-AR

Project Number: 1345-A-126



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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-S-63382 Safe and Arming (Partial Loading) Assembly for Mines, M718/M741 System

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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